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The palaeomagnetism of Late Permian–Early Triassic and Late Triassic deposits on Timor: an Australian origin?

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SUMMARY

Palaeomagnetic studies have been carried out on sediments from two formations in Western Timor; (a) red, crinoid-bearing limestones of the Maubisse Formation of Late Permian–Early Triassic age and (b) reddish calcilutites and red shales of the Aituto Formation of Late Triassic age. The crinoid-bearing limestones occur in both a non-recrystallized and a recrystallized, slightly marmorized facies. Natural Remanent Magnetization (NRM) was analysed by means of progressive demagnetization both using alternating magnetic fields and by heating. The determination of acquisition remanence and remanent coercive force demonstrated that in a large number of sites high-temperature (volcanic) haematite is the main carrier of remanence. A few sites from the non-recrystallized Maubisse Limestones revealed magnetite as the main remanence carrier. The recrystallized Maubisse Limestones were found to have a secondary remanence with a very high coercivity and a high blocking temperature. Characteristic Remanent Magnetization (ChRM) directions could be obtained both from the non-recrystallized limestones of the Maubisse Formation and from the Aituto Formation. For the plate tectonic interpretation of the data only the ChRM inclination values have been used, because of the complicated, often chaotic tectonism on Timor. Palaeolatitudinal positions of 39° and 33° have been derived from the Late Permian–Early Triassic Maubisse Formation and the Late Triassic Aituto Formation, respectively. This points to the presence on the island of Timor of a displaced terrane of Australian origin.

Key words: Banda Arc, Indonesia, palaeomagnetism, rock magnetism, Triassic.

INTRODUCTION

The island of Timor is located on the non-volcanic outer arc of the Sunda–Banda Arc System (Fig. 1). Timor is the easternmost island of the Lesser Sunda Archipelago, situated east of longitude 120 °E. At about this longitude the main system is divided into two segments: (a) a western segment including the islands of Sumatra and Java, representing a classic example of subduction where the Indian Ocean crust underthrusts the continental crust of the Sunda plate, leading to the development of an outer arc trough, a non-volcanic arc, a volcanic arc, and interarc basins; and (b) an eastern segment, including Timor, where the continental crust of the Australian plate collides with the Banda Arc system (Hamilton 1979).

Usually, in a subduction system the non-volcanic outer arc is built up of sediments scraped off from the descending and subducting ocean crust, and the outer arc trough is underlain by oceanic crust. However, the situation on and around Timor is different. The island is not built up exclusively of sediments detached from an oceanic crust originally located in front of the Australian continental shelf which is at present moving northward. Timor is made up of rocks from different areas (Barber 1981), and the island is underlain by a continental crust (Chamalaun & Grady 1978). Furthermore, the Timor Trough, which is about 3 km deep and is located to the south of the island, cannot be considered as a subduction trough. The Timor Trough, underlain by continental crust and containing shallow marine sediments of mainly Pliocene age, is a fairly recent area of subsidence (Johnston & Bowin 1981). The volcanic inner arc, situated to the north of the island of Timor, has been virtually inactive since Middle to Late Pliocene times (Barber 1981), indicating that at present there is no subduction in this area. This is confirmed by the fact that there are very few earthquakes in this region.

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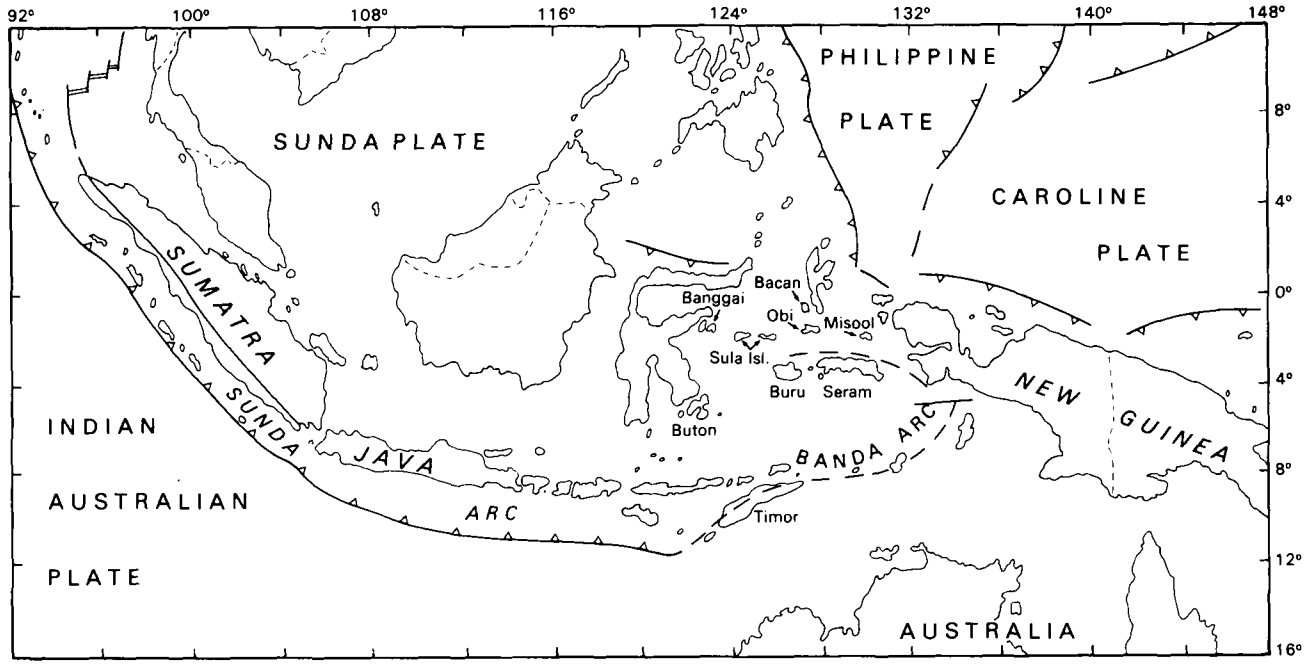


Figure 1. Map of the Indonesian Archipelago showing the main structural elements.

By Middle Pliocene times the frontal part of the Australian continental shelf had reached the Banda subduction zone. Then, subduction came to an end, and the volcanic activity in the greater part of the Banda System stopped. The island of Timor, which was very probably an old continental fragment (see below), was underthrust by the advancing frontal part of the Australian plate. Thus, the present edge of the plate consists of an Australian continental crust covered by rocks of the fragment, Timor. As a result, the Timor area was uplifted up, and an area to the south—the present Timor Trough—subsided.

In order to understand the structural history of the island of Timor we refer to the model of Pigram & Panggabean (1984). According to these authors, in Jurassic and Early Cretaceous time rifting took place along Australia's north and west coasts and subsequently, a northwestern corner separated from the main Australian plate and drifted towards SE Asia. This detached continental sub-plate broke up into a number of microcontinents which are now dispersed, and occur chiefly in Eastern Indonesia. Timor may be such a microcontinent which came originally from Australia. It is possible that Timor reached the Asian continent by the Late Mesozoic, broke away in the Early Tertiary and subsequently drifted southwards (Barber 1981). Therefore, it cannot be ruled out that Timor is built up of rocks of both Australian and Southeast Asian origin (Audley-Charles 1968; Barber, Audley-Charles & Carter 1977).

GEOLOGY OF TIMOR

General

Reviews of the geology of Timor have been presented by van Bemmelen (1949), Hamilton (1979) and Barber (1981). A geological map of Western Timor—with description—has been made by Rosidi, Suwitodirdjo & Tjokosapoetro (1979). It is generally recognized that the geology of Timor is very complicated. The oldest rocks are metamorphosed; they are exposed mainly in the northern part of the island, and are of pre-Permian age. Practically none of the younger sediments and volcanics on Timor underwent metamorphism. The island displays a great variety of rocks of possibly different areas of origin, viz. Australia and SE Asia; but there are also exposures with deep sea sediments, originally deposited both south (Wensink, Hartosukohardjo & Kool 1987) and north of Timor's present position.

On Timor, the majority of the rocks occur in rather a chaotic manner. An important formation on the island is the Bobonaro Scaly Clay (Audley-Charles 1968) or Bobonaro Complex (Rosidi *et al.* 1979). The very incompetent scaly clay of Miocene age forms a matrix in which exotic blocks of all dimensions are embedded, derived from older formations. The orientation of the blocks is fairly regular, sub-parallel to the main axis of the island. The Bobonaro Complex can be considered as an olistostrome, a submarine slump, comparable to the Argille Scagliose of the Italian Apennines (Merla 1951). The olistostrome probably moved from north to south. This sense of movement is not in conflict with the supposition that the Australian plate underthrusted the crustal fragment—Timor—causing the continental northern edge to be lifted up and giving rise to southward directed movements of the cover (Barber 1981; Johnston & Bowin 1981).

Sediments of Late Permian–Early Triassic and Late Triassic age

The present paper reports the results of a palaeomagnetic analysis of sediments of Late Permian–Early Triassic age of

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the Maubisse Formation, and of Late Triassic age of the Aituto Formation.

The Maubisse Formation is built up of well-bedded, red limestones, which are quite often exceptionally rich in debris of crinoids; one also finds brachiopods and ammonites. Usually, thin beds of calcareous shale are intercalated between the limestone layers. The type locality of the formation lies near the village of Maubisse in Eastern Timor, from where Audley-Charles (1968) reported the occurrence of ammonites and brachiopods indicating a Middle to Late Permian age and also identified corals pointing to a Late Permian age. Berry, Burrett & Banks (1984) studied the conodonts from identical red limestones of the Maubisse Formation from the same area in Eastern Timor. In their opinion, these conodonts refer to the *Neogondella milleri* biozone of the Upper Smithian Stage (Early Triassic). The well-preserved conodonts have a Colour Alteration Index of 1, suggesting regional metamorphic temperatures of less than 100 °C (Harris 1979). It is possible that the Maubisse Formation covers a rather large interval of time. We shall ascribe a Late Permian–Early Triassic age to these red, fossiliferous limestones.

The limestones with their fauna provide strong evidence that they were deposited in a warm shallow sea. Similar limestones occur on the northwestern Australian offshore shelf (Laws & Kraus 1974). The thickness of the formation is difficult to estimate, but it may be as much as 1000 m.

The limestones of the Maubisse Formation occur as a number of klippen of many different sizes. The outcrops sampled have dimensions of up to 1 km or more. The rocks of this formation make an important contribution to the exotic blocks of the Bobonaro Scaly Clay. In the eastern part of Western Timor the Maubisse Formation contains, in addition to non-recrystallized limestones, fairly strongly recrystallized, slightly marmorized red limestones.

The Aituto Formation consists of dense, multicoloured, fine grained, radiolarian calcilutites with a wide variety of thickness and bedding. There are intercalated red shales and marls; quartz-arenites occur as well (Rosidi *et al.* 1979). The type locality of the formation is in the Aituto anticline in Eastern Timor, where the thickness of the formation is estimated to be about 1000 m (Audley-Charles 1968). Usually the formation is not very fossiliferous; thin-shelled *Halobia*-type pelecypods occur frequently, pointing to a Carnian–Norian (Late Triassic) age for these sediments.

The marine deposits of the Aituto Formation were laid down under quiet conditions; however, the water depth at which sedimentation occurred is difficult to estimate.

The formation exhibits complicated structures. Again, one comes across structural units of many dimensions, very often surrounded by the Middle Miocene Bobonaro Scaly Clay.

PALAEOMAGNETIC SAMPLING

In the field we collected both oriented hand samples and oriented cores.

In the Maubisse Formation (Table 1) non-recrystallized limestones and recrystallized limestones were sampled. Oriented samples were collected at two outcrops of the non-recrystallized limestones: (a) eight sites in a number of small quarries, about 7 km east of Kefamenanu [TPAA–TPAH: locality (4) in Fig. 2]; (b) four sites near Suanae, 20 km SW of Kefamenanu [TPAI–TPAM: loc. (6) in Fig. 2].

Table 1. Palaeomagnetic localities with sampling sites in sediments of Late Permian–Early Triassic and Late Triassic age from Western Timor.

Formation	Sites	Locality	Position in Fig. 2
Maubisse (non-recrystallized)	TPAA - TPAH TPAI - TPAM	E of Kefamenanu near Suanae	(4) (6)
Maubisse (recrystallized)	TPLA - TPLI TPLK - TPLM TPLN - TPLP	Neonefo Kiumuna Kiukupan	(2) (5) (3)
Aituto	TPMA - TPMG TPMH - TPMK TPML - TPMN	Maubesi river Sabau river Heum river	(1) (7) (8)

Recrystallized limestones of the Maubisse Formation were sampled at three outcrops: (a) 9 sites near Neonefo, 27 km E of Kefamenanu, in a large exposure with steep E dipping well-bedded limestones [TPLA–TPLI: loc. (2) in Fig. 2]; (b) 3 sites near Kiumina, 4 km WNW of Kefamenanu [TPLK–TPLM: (5) in Fig. 2]; (c) three sites near Kiukupan, slightly east of the locality near Neonefo [TPLN–TPLP: (3) in Fig. 2].

From the Aituto Formation (Table 1) samples were collected at three localities: (a) seven sites in outcrops in the Maubesi river, 25 km east of Kefamenanu [TPMA–TPMG: loc. (1) in Fig. 2]; (b) three sites in the Sabau river, 20 km north of Soë [TPMH–TPMK: loc. (7) in Fig. 2]; and three sites in the Heum river, 5 km south of Soë [TPML–TPMN: loc. (8) in Fig. 2]. The lithology at the outcrops in the Heum river is of a different facies, and is made up of laminated sandstones and bituminous shales which are intercalated in the sequence of multicoloured limestones and red marls; here, burrowing structures were found.

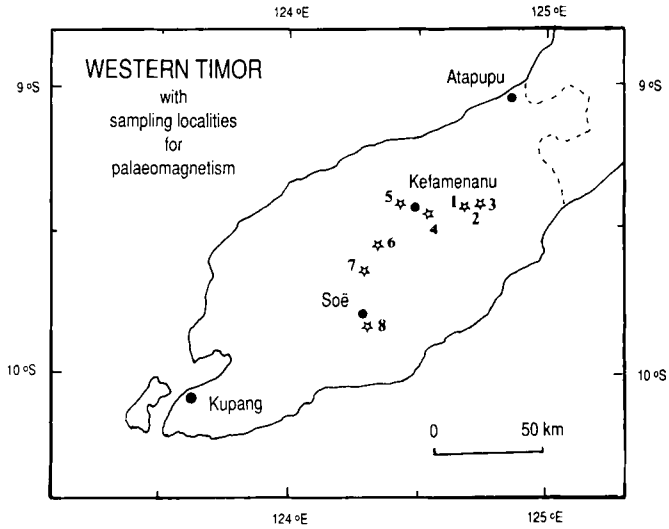


Figure 2. Map of Western Timor showing the palaeomagnetic sampling localities from two formations: (a) Maubisse Formation of Late Permian–Early Triassic age with non-recrystallized limestones near Kefamenanu (4), and near Suanae (6); and recrystallized limestones near Neonefo (2), near Kiumina (5), and near Kiukupan (3); (b) Aituto Formation of Late Triassic age in the Maubesi river (1), in the Sabau river (7), and in the Heum river, south of Soë (8).

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LABORATORY TREATMENT

General

The sampled material was prepared for palaeomagnetic analysis in the laboratory. Specimens used were cylindrical, with a diameter of 25 mm and a length of 22 mm.

Unfortunately, hand samples from the Aituto Formation, collected in the Maubesi river at three sites [TPMB, -E, and -G: Table 1, loc. (1) in Fig. 2], turned out to be too brittle for drilling cores in the laboratory.

All specimens were subjected to standard progressive demagnetization procedures both with alternating magnetic fields (AF) and with heating (thermal). The specimens were measured on a 2GT cryogenic magnetometer. To determine the directions of the components of remanence of a particular specimen, its demagnetization curves were plotted in orthogonal projection (Zijderveld 1967).

Non-recrystallized limestones of the Maubisse Formation (TPAA–TPAM)

The initial intensities (J_{rn}) of the Natural Remanent Magnetization NRM of specimens from the non-recrystallized limestones of the Maubisse Formation range from 0.3 to 42.4 mA m^{-2} (Table 2).

Progressive demagnetization was carried out both by heating and using alternating fields (AF). Progressive heating demonstrates that the initial NRM direction is usually built up of more than one component of remanence; generally, there is a viscous component which is destroyed by heating at low temperatures (Fig. 3: TPA K4). Several specimens appear to have three components of remanence: (a) a viscous component which is removed after heating between 110 °C and 250 °C; (b) a low blocking temperature component which disappears after heating in the temperature range between 325 °C and 375 °C; and (c) a characteristic component (Fig. 3: TPA B3; TPA E3). Usually, the remanance intensity of the characteristic component gradually decreases during progressive heating. After heating to 565 °C more than 20 per cent of the initial NRM intensity still remains. Very often, the characteristic component is still present after heating to 600 °C. Further heating occasionally resulted in an increase in remanence intensity, probably because of an alteration in the magnetic materials carrying the remanence; usually, this was accompanied by deviating remanence directions.

One specimen from each site was progressively demagnetized using the AF method, normally to a peak field of 75 mT (Fig. 3: TPA H1). AF treatment was not always successful: no characteristic component could be obtained for specimens showing a rapid decrease in the NRM intensity after treatment in low fields.

Recrystallized limestones of the Maubisse Formation (TPLA–TPLP)

The NRM initial intensities (J_{rn}) of specimens from the recrystallized limestones of the Maubisse Formation show a wide range. High J_{rn} values between 1.6 and 72.6 mA m^{-1} were observed in the Neonefo (TPLA–TPLI) and Kiukupan (TPLN–TPLP) outcrops (Tables 1 and 3). The material collected near Kiumina has much lower J_{rn} values, between 0.2 and 2.1 mA m^{-1}. The majority of the specimens were

Table 2. Palaeomagnetic data from non-recrystallized limestones of the Maubisse Formation of Late Permian–Early Triassic age, Western Timor.

Site	Attitude (Strike/Dip)	J_{rn}	N	E	af	Th	D	I	k	a_{95}
TPAA	222 / 11.5 NW *)	0.5 -1.6	7	1	1	7	288.8	-56.4	35	10.4
TPAB	263 / 21 N*)	0.4 -1.0	7	1	1	7	289.7	-54.2	43	9.3
TPAC	66 / 25 SE	3.8-15.5	4	6	1	9	267.9	-54.7	50	13.1
TPAD	222.5 / 14.5 NW*)	0.6 -1.1	10	-	1	9	268.6	-56.2	29	9.1
TPAE	125 / 20 SW	2.2-42.4	12	-	1	11	288.5	-5.6	2	39.1
TPAF	188 / 14 W	9.5-18.2	11	1	1	11	239.6	-55.0	7	18.1
TPAG	108 / 12 S	1.2 -1.5	7	-	1	6	290.8	-56.0	31	11.0
TPAH	121 / 5 SW	0.3 -1.2	10	-	1	9	274.0	-59.3	52	6.8
TPAI	107 / 12 S	0.4 -1.2	11	-	1	10	241.0	-74.4	15	12.2
TPAK	116 / 22 SW	0.6 -2.0	10	-	-	10	243.6	-66.9	65	6.0
TPAL	135 / 27 SW	0.4 -0.7	7	3	-	10	261.3	-50.1	7	25.0
TPAM	107 / 20 S*)	0.5 -1.0	9	1	-	10	258.4	-55.6	26	10.2

*Attitude is given by the strike and dip of the strata; J_{rn} is the initial remanence intensity in mA/m; N and E are the numbers of specimens included in, and excluded from the final analysis; af, Th, af+Th (Table 3) are treatments with alternating magnetic fields, heating, and alternating fields and subsequent heating, respectively; D and I are the declination and inclination in degrees of the characteristic remanence direction after bedding tilt correction; k is the precision parameter; a_{95} is the semi-angle of the cone of confidence, in degrees (Fisher, 1953); *) mean value.*

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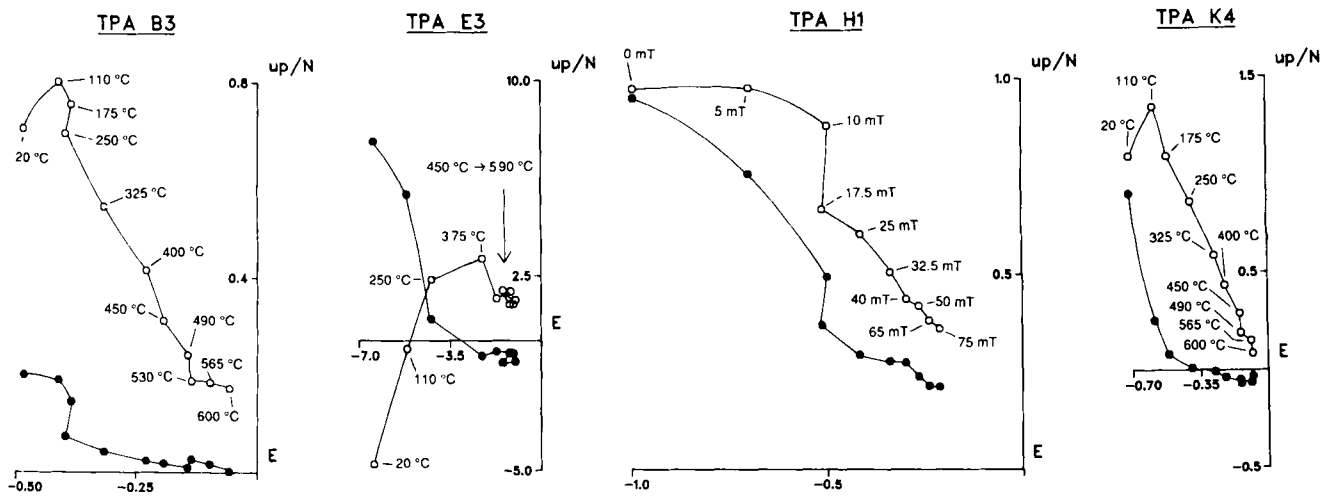


Figure 3. Diagrams showing the progressive demagnetization of specimens from non-recrystallized limestones of the Maubisse Formation (TPA sites). The plotted points represent successive positions in orthogonal projections of the end of the resultant remanence vector during progressive demagnetization. Some specimens were treated thermally, others were demagnetized with alternating fields. Solid and open circles denote the projections on a horizontal and a vertical plane, respectively. The units, indicated on both axes of the diagrams, are in mA m^{-1}.

subjected to progressive heating in 5–14 successive steps; the maximum temperature applied was 660 °C. Nearly all specimens from both the Neonefo sites (TPLA–TPLI) and the Kiukupan sites (TPLN–TPLP) had a viscous component of remanence of very low intensity which disappeared after heating to 105 °C (Fig. 4: TPL A5; TPL B6; TPL F4). The majority of the specimens from both localities revealed upon further progressive heating one single, stable component of remanence. An example of heating up to 645 °C is seen in Fig. 4: TPL B6.

The majority of the specimens from site TPLE, and some specimens from the sites TPLF through TPLI, contain not only the viscous and high-temperature component, but also an additional component of remanence (Fig. 4: TPL F4, two diagrams). Often, heating up to above 500 °C is needed to remove this additional remanence component.

The specimens from the Kiumina sites (TPLK–TPLM) show different behaviour during progressive heating: they reveal a rather large viscous component which hampers the determination of a characteristic component of magnetization.

One specimen from each of the TPL sites was

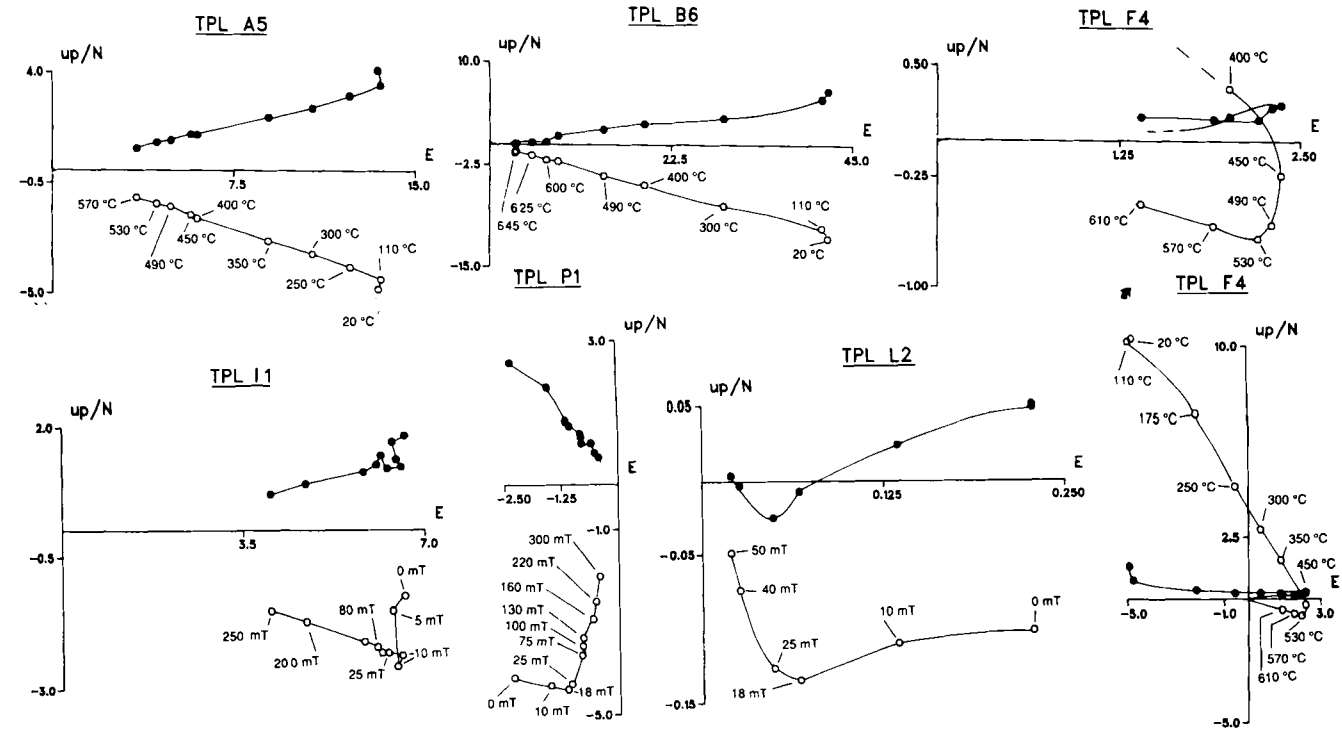


Figure 4. Diagrams showing the progressive demagnetization of specimens from the recrystallized limestones of the Maubisse Formation (TPL sites). Note Fig. 4 upper right shows a detailed diagram for TPL F4. See also caption to Fig. 3.

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Table 3. Palaeomagnetic data from red, recrystallized limestones of the Maubisse Formation of Late Permian–Early Triassic age, Western Timor.

Site	Strike/Dip	J_{rn}	N	E	af	Th	af+Th	D	I	k	a_{95}
TPLA	351 / 69 E	9.6 -42.9	6	1	-	6	1	81.5	21.0	327	3.7
TPLB	351 / 69 E	10.7 -49.1	9	-	-	8	1	85.0	16.2	801	1.8
TPLC	345 / 66 E	3.0 -15.0	10	-	1	9	-	80.7	18.7	543	2.1
TPLD	345 / 66 E	3.0 -33.7	9	-	-	8	1	80.0	20.1	504	2.3
TPLE	351 / 70 E	3.5 -65.5	8	2	1	9	-	85.9	12.9	184	4.1
TPLF	351 / 64 E	6.9 -72.6	9	1	-	9	1	84.4	18.0	382	2.6
TPLG	353 / 64 E	1.6 -58.9	11	1	1	11	-	83.8	18.7	325	2.5
TPLH	353 / 64 E	2.0 -17.5	5	5	-	9	1	89.1	15.6	327	4.2
TPLI	353 / 64 E	2.5 -21.4	8	-	-	7	1	79.5	11.8	100	5.6
TPLK	53 / 35 SE	0.3 -1.6	-	6	2	4	-	-	-		
TPLL	48 / 46 SE	0.2 -0.4	5	1	2	4	-	130.5	60.6	49	11.1
TPLM	76 / 44 S	0.3 -2.1	-	6	2	4	-	-	-		
TPLN	225 / 30 NW	39.3 -70.2	6	-	-	5	1	8.4	75.1	820	2.3
TPLO	212 / 40 NW	2.3 -11.1	7	-	1	6	-	336.0	77.3	187	4.4
TPLP	217 / 35 NW	5.4 -18.5	8	1	1	8	-	339.8	77.0	427	2.7

For explanation of symbols see Table 2.

demagnetized with progressive AF in up to 14 successive steps with maximum peak values between 75 mT and 250 mT. With AF treatment the specimens from the Neonefo sites (TPLA–TPLI) and the Kiukupan sites (TPLN–TPLP) reveal a very small viscous component, which disappears after treatment in fields of about 18 mT peak value, as well as a large characteristic component of fairly high coercivity (Fig. 4: TPL I1; TPL P1). During AF treatment the specimens from the Kiukupan sites (TPLK–TPLM) reveal much larger components of viscous remanence; these could be removed only after the application of peak fields of more than 25 mT (Fig. 4: TPL L2).

Red calcilutites and red shales of the Aituto Formation (TPMB–TPMN)

The specimens from the sites in the Maubesi river (TPMB–TPMG; see also Table 1) show a large variety of J_{rn} values: low J_{rn} values between 0.04 and 0.17 mA m^{-1} are reported from the sites TPMC and -D (Table 4). Palaeomagnetic analysis performed on specimens from these sites was not very successful (Table 4). The J_{rn} values of the specimens from sites TPMB and TPMF, which are between 0.22 to 0.73 mT m^{-1}, are somewhat higher. Much higher values could be determined from the material collected in the Sabau river (TPMH–TPMK) with J_{rn} of between 0.94

Table 4. Palaeomagnetic data from sediments of the Aituto Formation of Late Triassic age, Western Timor.

Site	Strike/Dip	J_m	N	E	D	I	k	a_{95}
TPMB	110 / 46 S	0.58 -0.73	10	-	300.0	-52.5	206	3.4
TPMC	108 / 50 S	0.04 -0.09	2	3	320.0	-74.7	-	-
TPMD	110 / 46 S	0.04 -0.17	7	2	270.9	-4.6	6	28.3
TPMF	75 / 26 S*	0.22 -0.66	8	2	293.6	-51.2	41	8.7
TPMH	340 / 84 E	1.98 -3.49	15	2	301.1	-51.1	31	7.0
TPMI	337 / 83 NE	0.94 -2.23	7	3	103.8	-59.5	34	10.5
TPMK	341 / 83 E	2.63 -3.43	11	-	281.7	-54.2	126	4.1
TPML	8 / 51 E*	0.68 -1.86	2	7	-	-	-	-
TPMM	**	0.21 -1.92	7	2	259.2	-31.9	56	8.2
TPMN	43 / 30 SE	0.57 -1.07	4	3	314.1	25.7	15	24.9

** mean value; ** site TPMM has several values for strike and dip with 10 / 88 E; 23 / 36 SE; and 51 / 55 SE; for further explanation see Table 2.*

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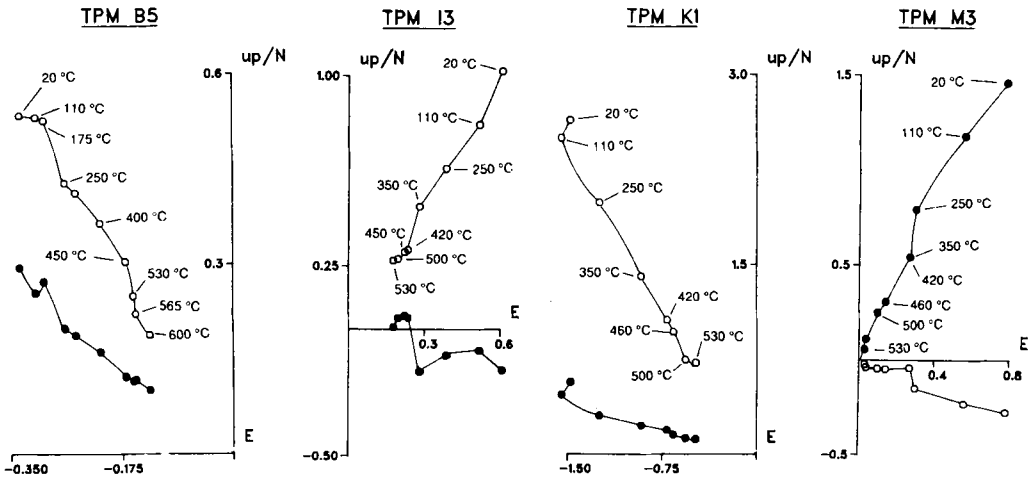


Figure 5. Diagrams showing the progressive demagnetization of specimens from the sediments of the Aituto formation (TPM sites). See also caption to Fig. 3.

and 3.49 mA m^{-1}. The J_{rn} values of the specimens from the sites in the Heum river, south of Soë (TPML–TPMN), range from 0.21 to 1.92 mA m^{-1} (Table 4).

Progressive heating only was applied to the specimens from the Aituto Formation. The majority of the specimens were subjected to heating in eight to 13 successive steps with maximum temperatures between 530 °C and 545 °C; some specimens were heated up to maximum temperatures above 600 °C. Many specimens from sites in the Maubesi river (TPMB–TPMF) and in the Sabau river (TPMH–TPMK) had a small viscous component of remanence that disappeared after heating at 110 °C (Fig. 5: TPM K1) or at 175 °C (Fig. 5: TPM B5). Upon further heating, specimens from these localities showed a gradual decrease in remanence intensity. Thus the majority of the specimens appear to carry only two components of remanence: a viscous component, and a characteristic component (Fig. 5: TPM K1); after heating above 530 °C the characteristic component still may be observable (Fig. 5: TPM B5). Very often, however, heating above 565 °C results in a rapid drop in remanence intensity combined with inconsistent magnetization directions.

During thermal treatment the specimens from site TPMI of the Sabau locality show a different behaviour (Fig. 5: TPM I3). After heating up to 420 °C the remanence intensity decreases, but upon subsequent heating up to 530 °C the intensity hardly diminishes any further; there is no change in remanance direction between 420 °C and 530 °C. After heating to still higher temperatures the magnetization directions become inconsistent. It is unlikely that any characteristic remanence direction can be determined from the specimens from site TPMI.

Progressive heating of specimens from sites in the Heum river, south of Soë (TPML–TPMN) was not successful. The

Table 5. Remanence Coercive Force (H_{cr}) and Isothermal Remanent Magnetization after application of direct field of 2 T (IRM_{2T}) in specimens from sediments of Late Permian–Early Triassic, and Late Triassic age, Western Timor.

Specimen	H_{cr}	IRM_{2T}	Specimen	H_{cr}	IRM_{2T}	Specimen	H_{cr}	IRM_{2T}
TPA A11	665	2.77	TPL C11	380	5.44	TPM B41	450	0.91
TPA B61	380	0.63	TPL D41	601	6.45	TPM C21	510	1.69
TPA C12	21	1.22	TPL E22	470	6.40	TPM D52	450	0.30
TPA D12	470	0.69	TPL F12	520	6.73	TPM F61	385	0.48
TPA E42*	19	1.97	TPL G61	550	3.12	TPM H6	240	2.47
TPA E62	27	0.75	TPL K42	475	0.26	TPM H8	280	2.10
TPA F12	23	4.41	TPL L31	610	0.18	TPM I4B	275	4.23
TPA H22	435	0.68	TPL M21	495	0.17	TPM K2A	295	2.06
TPA I32	720	1.60	TPL N61	380	4.92	TPM L5A	184	1.38
TPA K21	770	0.49	TPL O51	365	7.04	TPM M4A	150	0.21
TPA L12	317	0.74						
TPA M62	288	0.72						

*H_{CR} in mT; IRM_{2T} in A/m; * specimen TPA E42 has a high initial remanence intensity of 42 mA/m.*

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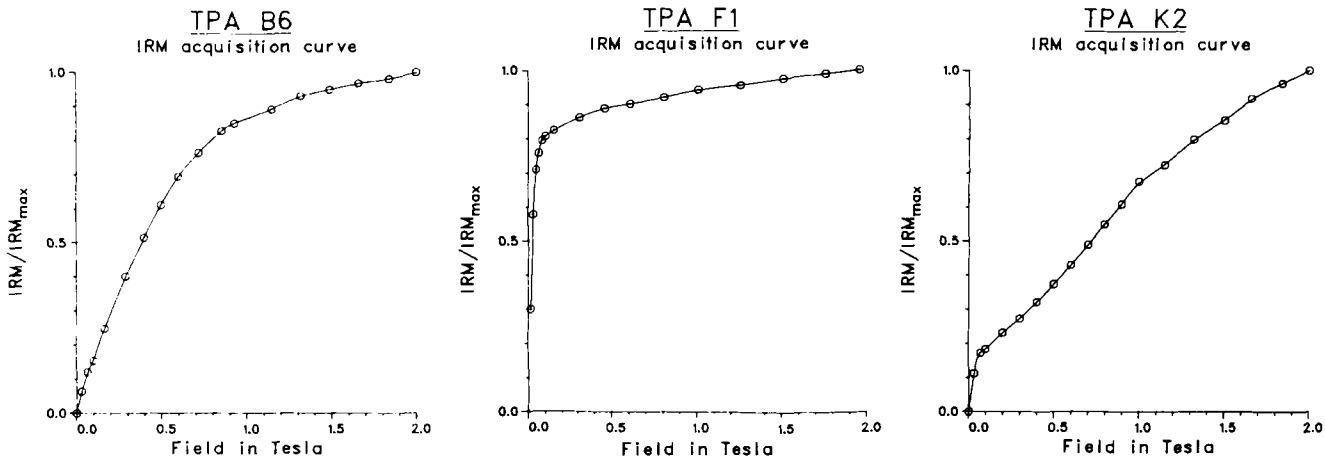


Figure 6. Normalized Acquisition curves of Isothermal Remanent Magnetization (IRM) for specimens from the non-recrystallized limestones of the Maubisse Formation; these point to the dominance of high coercivity mineral(s) in specimens TPA B6 and TPA K2, e.g. haematite, and a low coercivity mineral in specimen TPA F1, most probably magnetite.

demagnetization diagram of specimen TPM M3 (Fig. 5) shows the presence of one component only, which is virtually destroyed by heating at 530 °C.

ISOTHERMAL REMANENCE AND REMANENT COERCIVE FORCE

From one specimen from the majority of the sites we determined both the Isothermal Remanent Magnetization (IRM) after application of a direct field of 2 T (IRM_{2T}), and the Remanent Coercive Force (H_{cr}). The data are listed in Table 5. Both Normalized Remanence Acquisition curves and Normalized Remanence Hysteresis curves have been composed with a maximum direct field of 2 T (Figs 6, 7, 8, and 9).

In the red non-crystallized limestones of the Maubisse Formation, sampled near Kefamenanu (TPAA–TPAH), one can recognize two groups of sites with different values for H_{cr}: a first group of four sites (TPAA, -B, -D, -H) with high values between 380 and 665 mT; and a second group of three sites (TPAC, -E, -F) with low values between 21 and 27 mT (Table 4). In the first group haematite is an important carrier of natural remanence; this is supported both by the results of progressive heating (Fig. 3: TPA B3) and by the shape of the remanent acquisition curves (Fig. 6: TPA B6; Fig. 7: TPA A1, TPA H2). In the latter curves saturation remanence has still not been reached after the application of a direct field of 2 T. The second group has low H_{cr} values, indicating the possible presence of magnetite. The remanence acquisition curves show a rapid increase in the remanence intensity (J_{rem}) with increase of the applied direct field; this points to the presence of minerals of low coercivity, e.g. magnetite (Fig. 6: TPA F1; Fig. 7: TPA E4). The demagnetization diagrams indicate the prevalence of a mineral(s) with a fairly low blocking temperature(s) (Fig. 3: TPA E3); a mineral with a high coercive force, e.g. haematite, could be present as well (Fig. 6: TPA F1). Usually there is no clear relationship between the H_{cr} and the IRM_{2T} values; a small amount of magnetite may result in a relatively high IRM_{2T} value.

The non-recrystallized limestones of the Maubisse Formation, sampled near Suanae (TPAI–TPAM), reveal magnetic properties which permit a sub-division into two two groups: (a) a group with very high H_{cr} values of more than 700 mT (TPAI, -K); and (b) a group of two sites (TPAL, -M) with lower H_{cr} values (Table 5). The IRM acquisition curve for a specimen of the first group (Fig. 6: TPA K2) shows a very small, steep part, possibly indicating

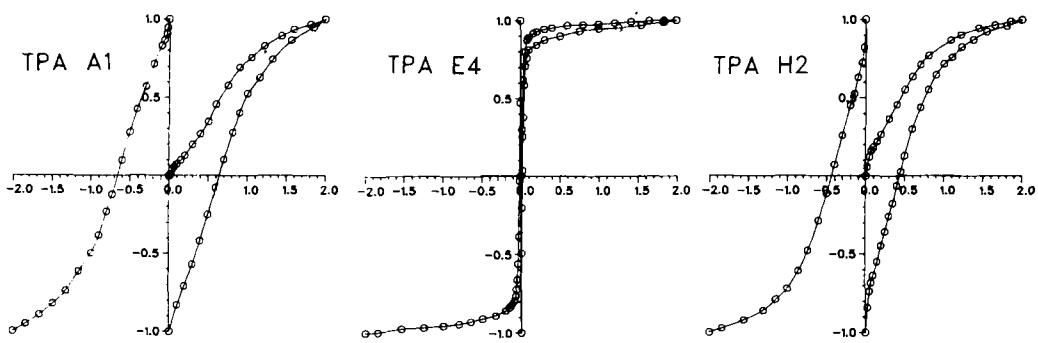


Figure 7. Normalized Remanent Hysteresis curves of specimens from the non-recrystallized limestones of the Maubisse Formation with the prevalence of (a) high coercivity mineral(s) in specimens TPA A1 and TPA H2, and (a) low coercivity mineral(s) in specimen TPA E4. The applied direct field is given in Tesla.

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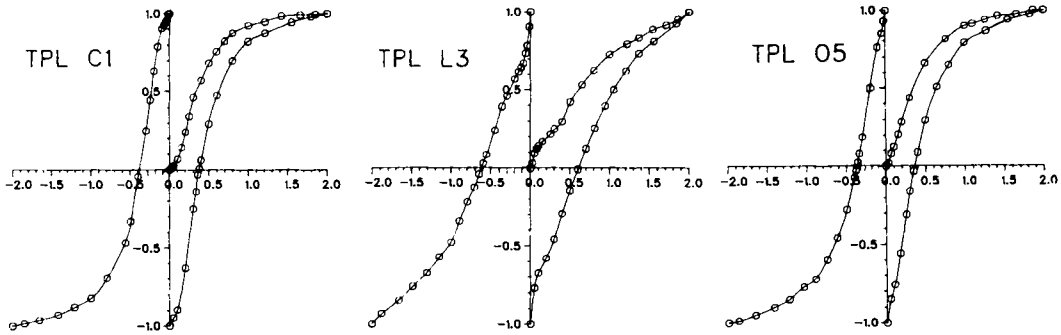


Figure 8. Normalized Remanence Hysteresis curves of specimens from the recrystallized limestones of the Maubisse Formation, indicating the prevalence of a carrier of remanence with a high coercivity, e.g. haematite. The applied direct field is given in Tesla.

magnetite, followed by a fairly constant increase at which saturation remanence has not been reached, even after application of the maximum direct field of 2 T. After progressive heating at 600 °C there is still a small proportion of the original remanence left (Fig. 3: TPA K4). It is possible that haematite with a very small grain size (less than 1 μm), and of high-temperature volcanic origin, is the main carrier of remanence (Hartstra 1982). In the second group haematite is also the main carrier of remanence; here, the presence of low-temperature haematite cannot be ruled out (Dekkers & Linssen 1989).

The red recrystallized limestones of the Maubisse Formation sampled at the Neonefo locality (TPLA–TPLI) and the Kiukupan locality (TPLN–TPLP) revealed fairly high H_{cr} values between 365 and 601 mT; the specimens from the Kiumina locality (TPLK–TPLM) have large H_{cr} values as well—between 475 and 610 mT—but much lower

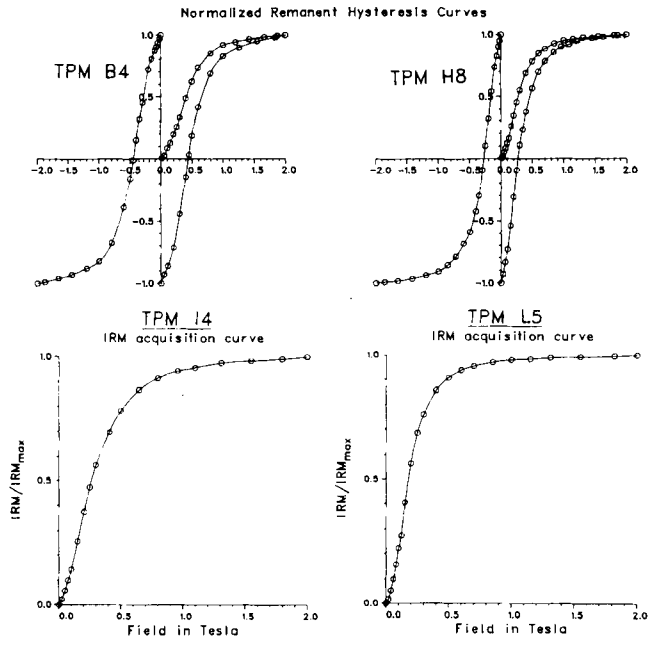


Figure 9. Normalized Remanence Hysteresis curves and Normalized Remanence Acquisition curves of IRM for specimens from sediments of the Aituto Formation. A mineral of high coercivity is the main carrier of remanence. The applied direct field is given in Tesla.

IRM_{2T} values, corresponding to relatively low J_{rn} values (Tables 3 and 5). Inspection of the remanence hysteresis curves of specimens representative of the first two localities reveals a fairly slow remanence increase with increasing direct field, indicating that a mineral of high coercivity, viz. haematite, is the predominant carrier of remanence (Fig. 8: TPL C1, TPL O5). Progressive heating of specimens from both localities shows the presence of a carrier of remanence with a fairly high range of blocking temperature (Fig. 4: TPL B6, TPL F4); progressive AF treatment points to the presence of a mineral with a high coercivity (Fig. 4: TPL I1, TPL P1).

IRM acquisition curves for the third locality (Kiumina: TPLK–TPLM) show the possible presence of a small amount of magnetite, as well as a remanence carrier of high coercivity, viz. haematite (Fig. 8: TPL L3). Progressive AF demagnetization points to the presence of a mineral of fairly low coercivity.

In the red recrystallized limestones it is difficult to establish the origin of the haematite. A study on low-temperature haematite (Dekkers & Linssen 1989) showed that its remanent coercive force has a maximum value of about 440 mT at a grain size of approximately 1 μm. The majority of the studied specimens from the red crystallized limestones could have high-temperature—possibly volcanic—haematite as a carrier of remanence.

The sediments from the Aituto Formation have various magnetic properties. Material from the Maubesi river (TPMB–TPMF) revealed fairly high H_{cr} values, between 385 and 510 mT; both the J_{rn} and the IRM_{2T} values are low (Tables 3 and 5). The shape of the IRM acquisition curves (Fig. 9: TPM B4) points to the presence of a carrier of remanence with a high coercivity. The demagnetization diagrams show the presence of a high blocking temperature component (Fig. 5: TPM B5). The main carrier of remanence is most probably haematite, part of which could have a high-temperature origin.

The Aituto sediments collected in the Sabau river (TPMH–TPMK) have H_{cr} values between 240 and 295 mT; both the J_{rn} values and the IRM_{2T} values are fairly high (Tables 4 and 5). The IRM acquisition curves (Fig. 9: TPM H8, TPM I4) indicate the possible presence of both a mineral of fairly low coercivity, e.g. magnetite (TPM H8), and a mineral with a high coercivity, e.g. haematite. No definite conclusions can be drawn from the demagnetization diagrams (Fig. 5: TPM I3, TPM K1).

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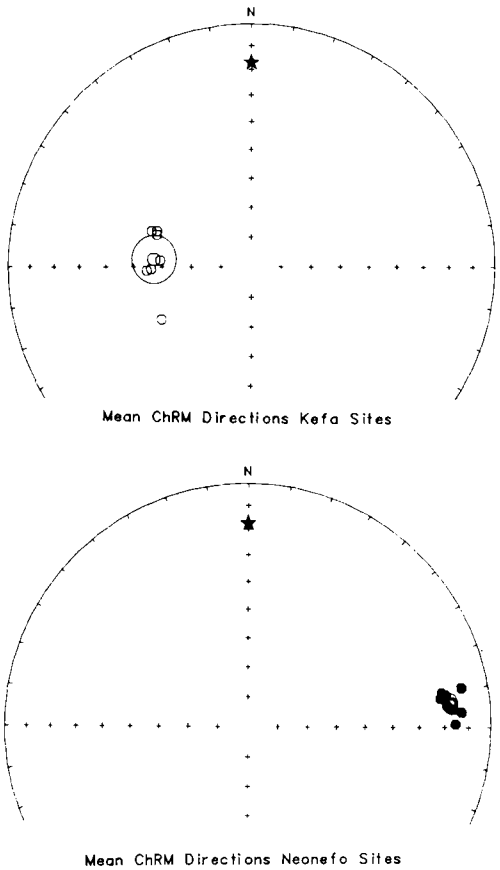


Figure 10. Stereoplots of the Characteristic Remanence (ChRM) directions at sites in the non-recrystallized limestones of the Maubisse Formation near Kefamenanu (TPAA–TPAH: upper plot), and in the recrystallized limestones of the Maubisse Formation near Neonefo (TPLA–TPLI: lower plot). 95 per cent confidence circles are shown. Dark and light symbols denote downward-pointing and upward-pointing directions, respectively. The star marks the local direction of the present axial geocentric dipole field.

The Aituto sediments from the Heum river (TPML–TPMN) revealed rather low H_{cr} values between 150 and 184 mT. The IRM acquisition curves (Fig. 9: TPM L5) show the prevalence of a remanence carrier of rather low coercivity; from the demagnetization diagrams one can see that a low blocking temperature component predominates. These sediments may contain both magnetite and a small amount of haematite.

The haematite in the sediments from the Sabau (TPMH–TPMK) and the Heum (TPML–TPMN) localities is not indisputably of high-temperature origin. It is plausible that low-temperature haematite with a grain size smaller than 0.5 μm is an important carrier of remanence.

PALAEOMAGNETIC RESULTS

The Maubisse Formation (non-recrystallized limestones)

The palaeomagnetic data—after tectonic correction—derived from the red, recrystallized limestones of the Maubisse Formation are listed in Table 2. Palaeomagnetic analysis of the specimens from both localities—near Kefamenanu (TPAA–TPAH) and Suanae (TPAI–TPAM)—provided consistent results. Careful progressive demagnetization resulted in the detection of a ChRM direction in the majority of the specimens from both localities (Fig. 10). Nevertheless, in some sites the within-site directions are rather dispersed. Site TPAE—showing a large value for a_{95}—revealed scattered ChRM directions. The combined palaeomagnetic data from the non-recrystallized Maubisse Formation are presented in Table 6; for both localities the data are given both before and after tectonic correction. It appears that the fold test is positive: after tectonic correction the a_{95} values are much lower.

There is a difference of about 20° between the ChRM declination values derived from both localities, indicating that the limestone units underwent independent rotations; the difference in the ChRM inclination values is rather small. The palaeomagnetic pole positions for the non-recrystallized limestones of the Maubisse Formation are listed in Table 7.

The Maubisse Formation (recrystallized limestones)

The palaeomagnetic data—after tectonic correction—derived from the material of the red, recrystallized limestones of the Maubisse Formation are listed in Table 3.

Table 6. Combined palaeomagnetic data from non-recrystallized limestones of the Maubisse Formation of Late Permian–Early Triassic age, Western Timor.

Locality	Sites	TC	N	D	I	k	a_{95}	Palaeolatitude
Kefamenanu	TPAA-TPAH	No	7	273.5	-51.6	20	13.8	-
Kefamenanu	TPAA-TPAH	Yes	7	274.4	-57.1	62	7.7	37.7
Suanae	TPAI-TPAH	No	4	237.2	-45.9	21	20.5	-
Suanae	TPAI-TPAM	Yes	4	253.8	-62.0	47	13.6	43.2
Both	TPAA-TPAM	Yes	11	267.6	-59.3	48	6.7	40.1
Both	TPAA-TPAM	Yes	11	-	-58.1*	76	4.2	38.8

*TC is tectonic correction applied; N is the number of sites included in the analysis; * mean inclination only (McFadden & Reid, 1982); for explanation of symbols see Table 2. Site TPAE from locality Kefamenanu is excluded from the final analysis.*

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Table 7. Palaeomagnetic pole positions for non-recrystallized limestones of the Maubisse Formation, Western Timor.

Locality	Coordinates		Pole Position			
			Lat. (°)	Long. (°)	dp (°)	dm (°)
Kefamenanau	9.48 °S	124.52 °E	9.2 S	177.6 E	8.2	11.2
Suanae	9.57 °S	124.35 °E	5.0 N	169.0 E	16.4	21.1

dp and dm are the semi-axes of the oval of 95 % confidence for the pole position.

The sites in the Neonefo locality (TPLA–TPLI), each showing small a_{95} values, yield consistent palaeomagnetic directions (Fig. 10). Consistent, but deviating directions also were obtained from the sites in the Kiukupan locality (TPLN–TPLP). Palaeomagnetic analysis of the material from the Kiumina locality (TPLK–TPLM) was not successful; no ChRM directions could be detected.

The combined palaeomagnetic data from the Neonefo and the Kiukupan localities are listed in Table 8 both after and before tectonic correction. The palaeomagnetic directions from both localities—after tectonic correction—differ considerably both in declination and in inclination values. It is not likely therefore that the ChRM directions detected from these recrystallized limestones are of primary origin. We have seen that in these limestones usually the initial NRM direction is composite and consists of two components: (a) a very small viscous component; and (b) a very stable magnetic component (Fig. 4). Table 8 shows that there is no agreement in the palaeomagnetic directions before the application of a tectonic correction either. One can conclude that, most likely, the recrystallized limetones of the Maubisse Formation have been strongly remagnetized during the process of deformation. At the time of remagnetization the Neonefo and Kiukupan limestone units may have had different tectonic positions; then, after their remagnetization the limestone units did not undergo the same amount of tilt.

Table 8. Combined palaeomagnetic data from recrystallized limestones of the Maubisse Formation of Late Permian–Early Triassic age, Western Timor.

Locality	Sites	TC	N	D	I	k	a_{95}
Neonefo	TPLA-TPLI	Yes	9	83.3	17.0	345	2.8
Kiukupan	TPLN-TPLO	Yes	3	349.0	76.9	347	6.6
Neonefo	TPLA-TPLI	No	9	103.0	82.3	243	3.3
Kiukupan	TPLN-TPLO	No	3	108.1	66.5	594	5.1

TC means tectonic correction applied; N is the number of sites included in the analysis; for explanation of symbols see also Table 2.

Table 10. Palaeomagnetic pole position for the Aituto Formation Western Timor.

Locality	Coordinates		Pole Position			
			Lat. (°) S	Long. (°) E	dp(°)	dm(°)
Maubesi	9.45 °S	124.65 °E	27.6	182.8	-	-
Sabau	9.67 °S	124.26 °E	23.4	181.8	-	-
Both	9.55 °S	124.45 °E	25.5	182.2	6.0	8.7

For explanation of symbols see Table 7.

The Aituto Formation

The palaeomagnetic data—after tectonic correction—derived from the sediments of the Aituto Formation are listed in Table 4. After a careful study of the material sampled in the Maubesi river (TPMB–TPMF) it was found that the data from two sites only were suitable for further palaeomagnetic analysis (Table 4: TPMB, -F). Palaeomagnetic research on the sediments collected in the Sabau river (TPMH–TPMK) was more successful. Corresponding ChRM directions could be obtained from sites TPMH and TPMK. However, deviating directions were found in specimens from site TPMI. The presence of a hard, secondary component of magnetization hampers the detection of the ChRM component in specimens from site TPMI (see also Fig. 3: TPM I3). The palaeomagnetic data derived from the sites in the Heum river (TPML–TPMN) are of rather poor quality; they are omitted from the final analysis.

For the Aituto Formation, the combined palaeomagnetic data are given in Table 9. The corresponding palaeomagnetic pole positions are listed in Table 10.

Table 9. Combined palaeomagnetic data from sediments of the Aituto Formation of Late Triassic age, Western Timor.

Locality	Sites	N	D	I	k	a_{95}	Palaeolatitude
Maubesi	TPMB, TPMF	2	296.8	-51.9	-	-	32.5
Sabau	TPMH, TPMK	2	291.7	-53.0	-	-	33.6
Both	TPMB, -F, -H, -K	4	294.3	-52.5	212	6.3	33.1
Both	TPMB, -F, -H, -K	4	-	-52.3*	1667	1.0	32.8

* *Mean inclination only (McFadden & Reid, 1982); for further explanation see also Tables 2 and 6.*

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DISCUSSION OF THE RESULTS

Progressive thermal demagnetization, determination of remanence coercive force, and the inspection of both IRM acquisition curves and remanence hysteresis curves of analysed specimens from sediments of Late Permian–Early Triassic and Late Triassic age from Timor lead us to the conclusion that haematite is the main carrier of remanence. Magnetite is the main carrier of remanence in the specimens from three sites only, viz. the red non-recrystallized limestones of the Maubesi Formation, sampled near Kefamenanu (TPAC, -E, -F). Several sites revealed high H_{cr} values, above 500 mT, indicating the possible presence of a high temperature haematite. It is not easy to determine the grain size of the carriers of remanence. The grain sizes of haematite are probably mainly small, less than 5 μm, especially those of haematite of high-temperature origin (Hartstra 1982).

The mean ChRM directions derived from two localities within the non-recrystallized Maubisse Limestones, near Kefamenanu (TPAA–TPAH) and Suanae (TPAI–TPAM), show a 20° difference in declination (Table 6). These localities lie about 20 km apart. In Timor, non-corresponding declination data of the ChRM direction can be expected for different localities of a particular formation, because the majority of the main rock units lie in a chaotic manner. Therefore, it is very difficult to use the declination data for further interpretation. The inclination data, however, permit us to establish the palaeolatitude of the sediments, i.e. the latitude at which the deposits were laid down. For the analysis of the palaeomagnetic inclination data the method introduced by McFadden & Reid (1982) has been used. The mean inclination of −58° corresponds to a palaeolatitude of 39°. The data are listed in Table 6.

We have demonstrated that the recrystallized limestones of the Maubisse Formation have very probably been remagnetized during tectonic positions different from the present ones.

The palaeomagnetic data from the Aituto Formation, derived from the localities Maubesi (TPMB, -F) and Sabau (TPMH, -K) lying about 50 km apart, show a difference of about only 5° in declination value (Table 9). The difference is very small, but could be accidental, because of the usually independent rotations of individual blocks. The palaeolatitude data are listed in Table 9; the mean inclination value of −53° corresponds to a palaeolatitude of 33°. Note the large value for k obtained by using McFadden & Reid's (1982) statistical method.

INTERPRETATION OF THE PALAEOMAGNETIC DATA

The mean ChRM directions derived from both non-recrystallized limestones of the Maubisse Formation and from sediments of the Aituto Formation have negative, upward directed inclinations. This implies that if the ChRM is depositional or early post-depositional, and was acquired in an area located in the magnetic southern hemisphere, the sediments were magnetized during a period of normal magnetic polarity. We shall demonstrate that the studied sediments may have a southern origin. The age of the Maubisse Formation is Late Permian–Early Triassic, possibly younger than the Kiaman Interval, a period of reversed magnetic polarity that lasted from the Late Carboniferous through the end of the Permian. During the Triassic time interval a large number of reversals of the earth's magnetic field occurred.

If there is a relation between the Permian and Triassic rocks in Timor and rocks of the same age in Australia, one can compare the palaeomagnetic data derived from both the Maubisse Formation and the Aituto Formation with the data from Australian rocks of the same age. However, one should realize that such a comparison may be of little value, because of the chaotic attitudes of the rock units on Timor. There are rather few, reliable Australian palaeomagnetic data for the Permian and Triassic epochs. For a comparison with the Late Permian–Early Triassic Maubisse Formation (Table 7) we have chosen an Australian palaeomagnetic pole at 47 °S, 165 °E (after Embleton & Schmidt 1977). The Maubisse sediments may have performed a large clockwise rotation with a mean value of about 55° relative to this Australian pole. The palaeomagnetic pole position derived from the sediments of the Aituto Formation of Late Triassic age (Table 10) can be compared with Embleton & Schmidt's (1977) Australian pole of Triassic–Jurassic age at 47 °S, 176 °E. Relative to this pole the Aituto sediments may have performed a clockwise rotation of 25°.

From the mean inclination of the ChRM direction of the non-recrystallized limestones of the Maubisse Formation we have derived a palaeolatitude of 39° (Table 6). In the palaeogeographic configuration of the southern continents of Smith, Hurley & Briden (1981) for the Late Permian–Early Triassic, the original position of the area of deposition of the Maubisse sediments lies slightly to the west of the present Australian northwest coast (Fig. 11). Because of the similarity between the Maubisse sediments and those found by drilling on the Australian continental platform it is quite possible that the Timor sediments originated in the southern hemisphere.

A palaeomagnetic study of the Cribas Formation of Late Permian age from Eastern Timor (Chamalaun 1977a) shows the palaeolatitudinal position of the sampling area to be 34°;

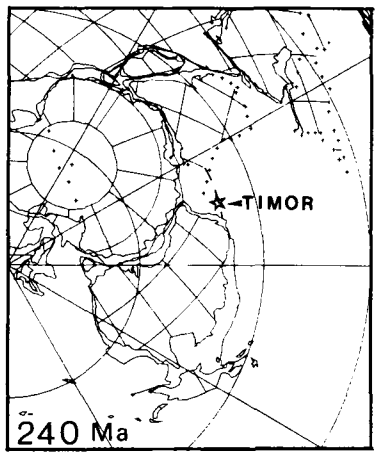


Figure 11. Map in Lambert equal area projection of the eastern part of Gondwana 240 Ma ago according to Smith *et al.* (1981) with the palaeolatitudinal position of the fragment of Timor at an arbitrary longitude during the deposition of the Late Permian–Early Triassic Maubisse limestones.

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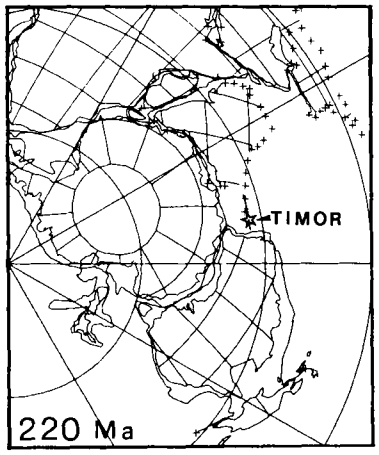


Figure 12. Map in Lambert equal area projection of the eastern part of Gondwana 220 Ma ago according to Smith *et al.* (1981) with the palaeolatitudinal position of the fragment of Timor during the deposition of the Late Triassic Aituto sediments.

the a_{95} derived from this formation is 7.1°. The mean magnetization direction derived from the Eastern Timor Maubisse Formation is $D = 140°$, $I = 45°$, with a value for a_{95} of 13° (Chamalaun 1977b); the corresponding palaeolatitude is 26°. Chamalaun argues that without interruption (Eastern) Timor formed part of the Australian continental margin. In our opinion, one cannot draw such far-reaching conclusions from Chamalaun's data.

The inclination of the mean ChRM direction of the Aituto deposits provides us with a palaeolatitude of 33° (Table 9). During the time interval Late Permian–Early Triassic to Late Triassic the Australian continent performed both a counterclockwise rotation and a shift towards the north. In the Late Triassic configuration of Eastern Gondwana, the depositional area of the Aituto sediments could have been located to the west of the Australian west coast (Fig. 12).

In conclusion it can be said that the palaeomagnetic research has demonstrated that rock units on the island of Timor, including sediments from both the Maubisse and the Aituto Formations, very probably originally formed part of the Australian continent. The complicated tectonics hamper the interpretation of the declination data of the ChRM directions derived from both formations. However, it is clear that both the Maubisse and the Aituto sediments performed rather large clockwise rotations relative to Australia. More research is needed if we are to understand the possible significance of these rotations. The interpretation of the palaeomagnetic data fits well into the pattern of structural evolution of northwest Australia, at which a microcontinent became detached in Jurassic to Early Cretaceous time. The present island of Timor contains fragments of Australian origin.

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